PA6 AI-use summary

This summary describes the AI assistance used to prepare the current PA6.m. Line references below refer to the 239-line version included with this summary.

How AI was used

- Lines 1-22: AI used the supplied MATLAB style example to organize a single script with clear, clc, close all, section headings, camelCase names, declared geometry, an adjustable query strain, and documented thresholds.
- Lines 24-86: AI wrote spreadsheet import and input checks, pre-rupture trimming, the 25-point force average, stress/strain conversion, and handling for small displacement reversals and exact repeats.
- Lines 88-124: AI selected an explicit estimation method for elastic modulus and proportional limit: fit the 5%-20% portion of the initial stress rise, then require a 5% departure for five consecutive points. These choices are adjustable assumptions, not independently verified material properties.
- Lines 126-152: AI implemented the 0.2% offset intersection using linear interpolation between adjacent samples, plus interpolation at the exact declared query strain. Out-of-range requests are not extrapolated.
- Lines 154-176: AI split the curve at the interpolated yield point, calculated resilience and toughness, and selected SI prefixes for output.
- Lines 177-239: AI wrote formatted command-window output, the curve plot, critical-point markers, labels outside the data, and red/cyan energy shading.
- AI wrote the README and this disclosure. The user supplied the assignment, example style, and instructions to use base MATLAB without add-ons and to perform static checks only.
- AI checked character vectors, delimiters, control-block nesting, base-MATLAB function usage, original input-file preservation, and Git whitespace errors. This program was not executed in MATLAB or Octave. Numerical results, graph appearance, and MATLAB Code Analyzer results remain unverified.
- This document does not claim that the student ran or personally verified the solution. Update it if the program, AI involvement, or validation changes.

Function explanations

Class and zyBooks coverage was not supplied. To support the assignment's disclosure rule, the following list explains the built-ins used, including ones that may already be familiar. All are part of base MATLAB.

- Lines 1-4, 206, 237-239 - clear, clc, close, warning, hold, grid, box: Clear workspace variables and command-window text; close figures; enable or report warnings; retain multiple plotted objects; and display plot grid lines and an axes border.
- Lines 21-22 - char: Convert character code 181 to the micro-prefix symbol for units. A cell array stores the SI prefix character vectors.
- Lines 24-43, 48-62, 79-99, 119-143 - error: Stop with an explanation when the input or identified elastic/critical-point region is unusable.
- Lines 27, 36-41, 79, 91, 98, 119, 132, 168-170, 186 - isscalar, isnan, isinf, all, any, size, numel, isempty: Check for a scalar query, missing numbers, and infinities; combine logical tests; count rows, columns, or elements; and check whether a search produced no index.

- Line 31 - readmatrix: Import the spreadsheet as a numeric matrix. Columns one and two supply displacement and force. This function requires MATLAB R2019a or later, without a toolbox.

- Lines 47, 83, 108, 172, 236 - max, min: Locate peak force and stress, protect a relative-deviation denominator, limit the SI exponent to the available prefixes, and choose axis limits.

- Line 68 - movmean: Take a centered 25-sample moving average of force. 'Endpoints', 'shrink' uses the available samples at the beginning and end; it does not pad with zeros or include post-rupture readings.

- Lines 75-78, 215 - sort, unique, accumarray, mean: Sort strain and reorder stress with the same indices. unique returns one copy of each strain and a group number for each reading. accumarray with @mean averages stresses in each exact-repeat group. @mean passes a built-in function handle, not an external library. A second sort orders point labels by stress.

- Lines 89-90, 129-131 - find: Locate the first index meeting the elastic-fit stress threshold or the offset-line sign-change condition. Empty results are checked before indexing.

- Lines 95-101 - polyfit, polyval: Fit a first-degree least-squares line to the initial elastic section and evaluate that line. Its slope is the estimated elastic modulus; the intercept helps assess proportionality. The offset-yield line separately passes through strain 0.002 at zero stress.

- Lines 107-108, 171-172 - abs, log10, floor: Measure an absolute deviation and choose a power-of-1000 SI scale. Rounding the base-10 exponent down to a multiple of three chooses a metric prefix.

- Line 149 - interp1: Linearly interpolate stress at queryStrain, rather than substituting the closest measured strain. The offset-yield intersection at lines 136-141 uses the equivalent two-point formula directly.

- Lines 159-160 - trapz: Integrate stress against the actual, nonuniform strain spacing using trapezoids. Pa times dimensionless strain gives J/m^3; division by one million gives MJ/m^3.

- Lines 166-167 - ones, cell: Allocate the numeric scale vector and a cell array of text prefixes before the output-formatting loop.

- Lines 178-200, 230 - fprintf, sprintf: Print aligned values with controlled decimal places or build a plot label containing the selected stress unit. %.4f prints strain to four decimals; %.2f prints stress to hundredths in its selected metric unit.

- Lines 205-211 - figure, area, plot: Create a figure; shade the energy areas to and after yield; and draw the smoothed stress-strain curve. Stored graphics handles control the legend's membership and order.

- Lines 217-227 - linspace, plot, text: Space four label heights evenly, draw critical-point markers and connecting lines, and position labels to the right of the measured curve so text does not cover its points.

- Lines 229-236 - xlabel, ylabel, title, legend, xlim, ylim: Set axis labels, title, legend entries and location, and visible plot limits. Disabling the unit label's text interpreter displays the prefix literally.

Assumptions to review before submission

- Lines 14-18, 45-60, 73-78, 88-124: Automatic elastic-region, proportional-limit, and rupture choices are heuristics for monotonic ductile tensile data. Large unloading, unusual noise, or a different sampling density can require adjusting the declared thresholds. Small strain reversals are treated as sensor jitter. The final sample is assumed to be fracture when a sudden post-peak force drop is absent.
- Lines 154-161: The energy calculation adds the origin, uses the interpolated yield point as the common boundary, and defines resilience as area to yield and toughness as area to fracture.
- The PDF is generated from this Markdown disclosure. Keep the line references and function explanations synchronized with any changes to PA6.m.